

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist who turns complex physical models into fast, tested software and turns large scientific datasets into calibrated models. Ph.D. in Physics and ten years building simulation codes and the data pipelines around them: author of Paramo, an open-source Fortran/OpenMP radiative-transfer library used by other research groups; led a 7-person upgrade of the PatchworkMHD black-hole simulation code on the Frontera supercomputer; mapped wetland probability across a 2,160 km² watershed with random-forest models. Nine peer-reviewed articles and Co-I on a funded NASA Fermi Guest Investigator grant.

Skills

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, random forest, SVM, neural networks, gradient-based optimization, parameter estimation, uncertainty quantification, synthetic dataset generation

Scientific Computing & HPC: Numerical PDEs (Fokker–Planck, advection–diffusion), radiative transfer, relativistic MHD, Lagrangian particle dynamics, finite-difference/finite-volume methods, numerical linear algebra, OpenMP, MPI, OpenACC, SLURM, PBS, HDF5

Geospatial & Remote Sensing: Sentinel-1/2 multispectral imagery, DEM terrain analysis, raster/vector pipelines, QGIS, geospatial feature engineering

Languages: Python, C/C++, Fortran, Rust, R, SQL, Java, Julia, Shell

Tools: Git, Docker, Azure, Jenkins, OpenShift, Kafka, Splunk, Jira, ParaView, VisIt

Experience

Independent Contractor

Jan 2026 – Present

Mercor

Remote

Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models and co-developed others with fellow writers.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.
- Benchmark model performance and quality-check solutions to complex STEM problems.

Spatial Data Scientist

May 2025 – Jan 2026

TealWaters

Seattle, WA

Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models; built Python processing tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer

Jan 2024 – May 2025

Self-directed

Seattle, WA

Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker–Planck equation coupled to radiative transfer; published in ApJ (2024) [9].
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Invited talk on machine learning in environmental science at the University of Washington (2024).

Software Engineer

Apr 2022 – Jan 2024

Paychex

Remote

Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts that flagged

production incidents.

- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.
- Prepared, validated, and analyzed Oracle EBS data with SQL.

Postdoctoral Research Associate

Rochester Institute of Technology

Feb 2021 – Apr 2022

Rochester, NY

HPC simulations of supermassive black-hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black-hole binaries.
- Added black-hole spin to the simulations with under 5% runtime overhead.
- Designed and ran benchmark simulations of the new features on TACC's Frontera supercomputer (about 1,200 node-hours), identifying performance optimization opportunities.
- Calibrated Paramo model parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, and supervised a graduate student applying gradient descent to the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron-star mergers [6, 7] and a third on blazar heating and cooling [8].

Postdoctoral Research Fellow

Purdue University

Oct 2018 – Nov 2020

West Lafayette, IN

Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Architected Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation coupled to radiative transfer; parallelized and optimized it with OpenMP, making it about 20× faster than the original serial code on 5 cores.
- Paramo has been adopted by other research groups, including in Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded (\$68K) and supported the finding that blazar jets are launched with similar energy per baryon, independently of their power [5].
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows [4].
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Wrote Python scripts for statistical modeling of the COVID-19 outbreak in Mexico with a multidisciplinary group, and co-wrote Spanish-language infographics and blog posts against misinformation.

Postdoctoral Research Fellow

Universidad Michoacana de San Nicolás de Hidalgo

Jan 2018 – Sep 2018

Morelia, Mexico

- Built a Python pipeline that turned GRTrans black-hole simulation images into training data for an SVM classifier of black-hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for computing radiative-transfer spectra and light curves in relativistic astrophysics.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant

Universitat de València

Oct 2011 – Jul 2017

Valencia, Spain

- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Analyzed multi-wavelength observations from the NASA Fermi-LAT telescope and the Very Long Baseline Array (VLBA), including cross-matching sources across catalogs by sky coordinates and fitting log-parabola spectra.
- Fit C-SPEV simulation models to telescope data, identifying spectral and light-curve patterns that quantify the magnetization of plasma in blazars.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS [2, 3].

Education

Ph.D. in Physics

Universitat de València

Oct 2011 – Jul 2017

Valencia, Spain

Excellent *cum laude*

Advisors: Prof. Miguel A. Aloy and Dr. Petar Mimica

Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

M.Sc. in Physics

Universidad Michoacana de San Nicolás de Hidalgo

Aug 2009 – Sep 2011

Morelia, Mexico

Advisor: Prof. José A. Cervera

Thesis: *Study of TOV stars with the SPH method*

B.Sc. in Physics

Universidad Autónoma del Estado de México

Aug 2004 – Dec 2008

Toluca, Mexico

Advisor: Prof. Francisco S. Guzmán

Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Lic. Juan Josafat Pichardo Cruz Award

Publications

Articles

- [12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C., *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.
- [11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C., *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.
- [10] **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ **976**, 182, (2024), arXiv:2405.17581.
- [8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS **513**, 5766–5779, (2022), arXiv:2201.07790.
- [7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D **106**, 083015, (2022), arXiv:2112.09817
- [6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ **919**, 95, (2021), arXiv:2106.05356
- [5] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS **501**, 4092–4102, (2021), arXiv:2009.02273.
- [4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS **496**, 974–986, (2020), arXiv:1910.14049.
- [3] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS **468**, 1169–1182, (2017), arXiv:1612.06383.
- [2] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS **438**, 1856–1869 (2014), arXiv:1310.5441.
- [1] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D **80**, 084023 (2009), arXiv:1009.1250.

Proceedings

5. **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 277–282, (2021), arXiv:2011.13797.
4. **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *The blazar sequence revised*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 147–152, (2021), arXiv:2011.13805.
3. **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) **233**, 159 (2014), arXiv:1502.07882.
2. **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A. & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. **61**, 02007 (2013), arXiv:1309.4612.
1. Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S. & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser **454**, 012001 (2013), arXiv:1211.1794.

Grants and Fellowships

Research Grants

NASA Fermi Guest Investigator Program, Cycle 12

A simple model to understand the blazar sequence

PI: D. Giannios; Co-I: J. M. Rueda-Becerril

2019

Grant #121077, \$68K

Fellowships

International Postdoctoral Fellowship <i>National Council of Science and Technology (CONACyT), Mexico</i>	Oct 2018 – Nov 2020
Postdoctoral Fellowship <i>Secretariat of Public Education, Mexico (Program for the Professional Development of Higher Education Institutions)</i>	Jan 2018 – Sep 2018
Study-Abroad Fellowship <i>National Council of Science and Technology (CONACyT), Mexico</i>	Sep 2014 – Aug 2016
Santiago Grisolia Fellowship <i>Council of Education, Research, Culture and Sport, Valencian Community, Spain</i>	Oct 2011 – Jun 2014
M.Sc. Fellowship <i>CONACyT, at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo</i>	Sep 2009 – Aug 2011
Summer Research Fellowship (XVII Summer of Scientific Research) <i>Mexican Academy of Sciences</i>	Jun 2007 – Aug 2007

Software and Projects

Blazar Population ML <i>Scientific ML (Python, scikit-learn, PyTorch)</i>	Jan 2026 – Present <i>github.com/altjerue/ilwikak-ml</i>
- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data. - Building a neural network that maps physical jet parameters to observable spectral energy distributions accurately enough to compare with Tleco results. - Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.	
WindsOfChange <i>Lagrangian particle-dispersion library (C++/R)</i>	Aug 2024 – Present <i>github.com/UW-ForestMycobiomeLab/WindSporesHills</i>
Simulates particle dispersion, atmospheric turbulence, and flow over hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004); designed to scale to tens of thousands of particles over kilometer-scale domains.	
Tleco <i>Radiative-transfer toolkit (Rust/Python)</i>	Jan 2024 – Sep 2024 <i>github.com/zkdavis/Tleco</i>
Numerical PDE solvers, particle-distribution algorithms, and spectral calculations for relativistic plasmas; successor to Paramo [9].	
Paramo <i>Radiative-transfer library (Fortran 95, OpenMP)</i>	Oct 2018 – Apr 2022 <i>github.com/altjerue/paramo</i>
Fokker–Planck and radiative-transfer solver for relativistic particle distributions; about 20× faster than the original serial code on 5 cores after OpenMP parallelization and optimization; used by other groups, including Combi & Siegel (2023).	
PatchworkMHD <i>Large-scale MHD simulation code (NSF project)</i>	Feb 2021 – Apr 2022
Led a 7-person upgrade of this production C code for HPC environments, adding black-hole spin with under 5% runtime overhead.	

Awards

Marcos Moshinsky Award for Best Poster <i>9th International Workshop on Astronomy and Relativistic Astrophysics (IWARA 2020), video conference</i>	Sep 2020
Lic. Juan Josafat Pichardo Cruz Award <i>Universidad Autónoma del Estado de México</i> For defending a licentiate thesis within a year of completing the undergraduate coursework.	2009

Presentations

Invited Talks

The Role of Machine Learning in Environmental Science <i>Seminar on Climate Change Ecology, University of Washington</i>	Mar 2024 <i>Seattle, WA</i>
Morphology of the spectra from numerical simulations of the internal shocks model for blazars <i>Astrophysics Seminar, Purdue University</i>	Feb 2019 <i>West Lafayette, IN</i>
Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars <i>DATA Group Weekly Seminar, Instituto de Astronomía, UNAM</i>	Jun 2018 <i>Mexico City, Mexico</i>
Numerical treatment of non-thermal radiation in the internal shocks model for blazars <i>Weekly Seminar, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo</i>	Mar 2018 <i>Morelia, Mexico</i>

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars <i>IVICFA's Fridays: Computation in Physics, IFIC</i>	Oct 2014 <i>Paterna, Spain</i>
Contributed Talks	
Simulations of supermassive binary black holes accretion dynamics in the spinning case <i>APS April Meeting 2022</i>	Apr 2022 <i>New York, NY</i>
A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows <i>APS April Meeting 2021</i>	Apr 2021 <i>Online</i>
The blazar sequence revised <i>9th International Workshop on Astronomy and Relativistic Astrophysics</i> Video: youtube.com/watch?v=BAZNWLNT69M	Sep 2020 <i>Online</i>
Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario <i>Extreme-Astrophysics in an Ever-Changing Universe</i>	Jun 2014 <i>Ierápetra, Greece</i>
Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets <i>XXXIV Biennial Meeting of the Royal Spanish Society of Physics</i>	Jul 2013 <i>Valencia, Spain</i>
Posters	
A numerical approach for radiative cooling in relativistic outflows <i>9th International Workshop on Astronomy and Relativistic Astrophysics</i> Marcos Moshinsky Award for Best Poster. Video: youtube.com/watch?v=0TJiKg7kOPI	Sep 2020 <i>Online</i>
Numerical simulations of the internal shock model in magnetized relativistic jets of blazars <i>Swift: 10 Years of Discovery</i>	Dec 2014 <i>Rome, Italy</i>
Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets <i>The Innermost Regions of Relativistic Jets and Their Magnetic Fields</i>	Jun 2013 <i>Granada, Spain</i>

Teaching and Mentoring

Zachary Davis, graduate student <i>Department of Physics and Astronomy, Purdue University</i> Co-authored papers [8, 9]	2018 – 2022 <i>Mentoring</i>
Amanda O. Harrison, graduate student <i>Department of Physics and Astronomy, Purdue University</i> Co-authored paper [5]	2018 – 2020 <i>Mentoring</i>
Hao Zhang, graduate student <i>Department of Physics and Astronomy, Purdue University</i> Co-authored paper [4]	2018 – 2019 <i>Mentoring</i>
Thermodynamics (graduate level) <i>Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo</i>	Jun 2018 <i>Substitute Instructor</i>

Professional Development

High Performance Computing on Frontera <i>Jason Allison et al., TACC, Austin, TX</i>	May – Jun 2021 <i>Lecture series</i>
Writing Winning Grants <i>Dr. Lauren Broyles, Purdue University</i>	Nov 2019 <i>Lecture</i>
XSEDE HPC Workshop: Summer Boot Camp <i>John Urbanic, Purdue University</i>	Jun 2019 <i>Workshop</i>
Data Analysis and Machine Learning with Python <i>Dr. Alejandro Torres, Universitat de València</i>	Feb 2017 <i>Workshop</i>
Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit <i>Dr. Vassilios Mewes, Universitat de València</i>	Jul 2016 <i>Workshop, 8 hours</i>
The Universe in the Light of PLANCK and BICEP2 <i>Prof. Nick Mavromatos, Universitat de València</i>	May 2014 <i>Lecture series, 2 credits</i>
Dark Matter <i>Prof. Alejandro Ibarra, Universitat de València</i>	Sep 2013 <i>Lecture series, 2 credits</i>
International Cargèse School on Cosmic Accelerators <i>Institut d'Études Scientifiques de Cargèse, France</i>	Apr – May 2013 <i>Summer school</i>

Introduction to C++ Programming

Dr. Jacek Generowicz, Universitat de València

Apr 2012

Workshop, 6 credits

Numerical Relativistic Astrophysics

Prof. Luciano Rezzolla, Universitat de València

Mar – Apr 2012

Lecture series, 9 hours

Fortran for Scientific Computing

HLRS, University of Stuttgart, Germany

Mar 2012

Workshop, 33 hours

Certifications

Gen AI Intensive

Generative AI

Kaggle

Data Scientist Professional with Python

Python programming, data science, data communication, machine learning

DataCamp

Machine Learning Scientist with Python

Machine learning, NLP, deep learning, image processing, big data

DataCamp

Mathematical Foundations of Machine Learning

Python programming, data science, statistics, machine learning

Udemy

Python for Statistical Analysis

Python programming, data science, data communication, machine learning

Udemy

Outreach

Los más rápidos y los más furiosos (The Fastest and the Most Furious)

Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco

Sep 2020

Online talk

Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics)

Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco

Aug 2020

Online talk

ANITA y la teoría de los universos paralelos (ANITA and the Theory of Parallel Universes)

Científicos Mexicanos en el Extranjero

mexiciencia.github.io/post/anita

May 2020

Blog post

¿Qué es el modelo SIR? (What is the SIR Model?)

Científicos Mexicanos en el Extranjero

mexiciencia.github.io/post/modelo-sir

May 2020

Blog post

Evolución del brote epidémico de COVID-19 (Evolution of the COVID-19 Epidemic Outbreak)

Científicos Mexicanos en el Extranjero

Data analysis and modeling collaborator. mexiciencia.github.io/post/covid19

Apr 2020

Blog post

Annual Department of Physics and Astronomy Poster Event

Purdue University

Nov 2019

3 posters

Post-Doc Panel Q&A: What Happens When We Complete Our PhDs?

Department of Physics and Astronomy, Purdue University

Apr 2019

Panelist

Annual Department of Physics and Astronomy Poster Event

Purdue University

Nov 2018

Poster

¿Decía Einstein la verdad? (Was Einstein Telling the Truth?)

Facultad de Ciencias, Universidad Autónoma del Estado de México

Mar 2009

Talk

Service

Organizing contributor

X Scientific Meeting of the Spanish Astronomical Society

Dec 2012

Valencia, Spain

Student representative, Governing Council of the Faculty of Sciences

Universidad Autónoma del Estado de México

Aug 2007 – May 2009

Toluca, Mexico

Languages

Spanish (native), English (full professional proficiency), Catalan (basic), Nahuatl (basic)